**Chemistry • Heat, activation energy and biological energy**

# Energetics

Official exercise • Complete solved answers • Original energy profiles

**Chapter 5 • Complete official exercise • 10 MCQs + 5 short + 3 constructed + 5 descriptive + 2 investigative**

### What this chapter is about

• **Enthalpy change** is $\Delta H = H_{products} - H_{reactants}$. Exothermic reactions release heat and have $\Delta H < 0$; endothermic reactions absorb heat and have $\Delta H > 0$.
• **Bond breaking** requires energy; **bond formation** releases energy. The net balance determines the reaction's enthalpy change.
• **Activation energy** is the barrier to reach the transition state. A large barrier can prevent a thermodynamically favourable reaction from occurring at a noticeable rate.
• **Respiration** releases usable biological energy. Glycogen provides short-term glucose storage, while lipids provide concentrated long-term energy storage.

**Exercise coverage:** All five official sections are solved in book order. Reaction profiles and the two bond-energy numericals were checked against the rendered exercise.

## A. Multiple Choice Questions

**1. The reaction $H_2 + Cl_2 \rightarrow 2HCl$ in sunlight is exothermic. From where does the energy come to break the bonds of $H_2$ and $Cl_2$?**

(a) by collisions between the molecules
(b) from sunlight
(c) from the surrounding
(d) by collisions of the molecules with the walls of the container

**Answer: (b) from sunlight** — The equation explicitly shows sunlight supplying the initial energy needed to disrupt the reactant bonds and initiate the reaction.

**2. Which reaction has the least value of activation energy?**

(a) $H_2(g) + ½O_2(g) \rightarrow H_2O(g)$
(b) $C(s) + O_2(g) \rightarrow CO_2(g)$
(c) $NaCl(aq) + AgNO_3(aq) \rightarrow AgCl(s) + NaNO_3(aq)$
(d) $H_2(g) + I_2(s) \rightarrow 2HI(g)$

**Answer: (c) $NaCl(aq) + AgNO_3(aq) \rightarrow AgCl(s) + NaNO_3(aq)$** — The aqueous precipitation reaction is a rapid ionic combination with a very small activation barrier; the other reactions require substantial bond disruption or initiation.

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**3. Formation of which hydrogen halide from the elements is an endothermic reaction?**

(a) HCl
(b) HF
(c) HBr
(d) HI

**Answer: (d) HI** — Formation of HI from $H_2$ and $I_2$ is endothermic; formation of HF, HCl and HBr is exothermic.

**4. What are the products of anaerobic respiration?**

(a) ATP + $CO_2$ + $H_2O$
(b) $CO_2$ + $H_2O$
(c) ATP + ethanol + $H_2O$
(d) ethanol + $H_2O$

**Answer: (c) ATP + ethanol + $H_2O$** — Option (c) is intended because anaerobic fermentation forms ethanol and energy (ATP). Its printed $H_2O$ is a source error: the chapter's own description also produces $CO_2$, not water.

**5. Which reaction do you expect to be a reversible reaction?**

(a) profile (a): products at much lower energy than reactants
(b) profile (b): products at slightly lower energy than reactants
(c) profile (c): products at moderately higher energy than reactants
(d) profile (d): products at much higher energy than reactants

**Answer: (b) profile (b): products at slightly lower energy than reactants** — Profile (b) has the smallest energy difference between reactants and products, so forward and reverse changes are comparatively feasible.

**6. What does it show when a chemical reaction is exothermic?**

(a) the bonds which break are weaker than those formed
(b) the bonds which break are stronger than those formed
(c) exothermic nature is not concerned with bond formation or bond breakage
(d) the reactants are more stable than the products

**Answer: (a) the bonds which break are weaker than those formed** — Less energy is needed to break weaker reactant bonds than is released when stronger product bonds form, so the net enthalpy change is negative.

**7. When NaOH and HCl are mixed, the temperature increases. The reaction is:**

(a) endothermic with a positive enthalpy change
(b) endothermic with a negative enthalpy change
(c) exothermic with a positive enthalpy change
(d) exothermic with a negative enthalpy change

**Answer: (d) exothermic with a negative enthalpy change** — Neutralization releases heat to the surroundings, so it is exothermic and ΔH is negative.

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**8. The average bond dissociation energy for the C–H bond is 412 kJ mol$^{-1}$. Which process will have enthalpy change close to 412 kJ mol$^{-1}$?**

(a) $CH_4(g) \rightarrow C(g) + 2H_2(g)$
(b) $CH_4(g) \rightarrow C(g) + 2H_2(g)$
(c) $CH_4(g) \rightarrow C(g) + 4H(g)$
(d) $CH_4(g) \rightarrow CH_3(g) + H(g)$

**Answer: (d) $CH_4(g) \rightarrow CH_3(g) + H(g)$** — Only option (d) breaks one C–H bond. Complete atomization in (c) breaks four C–H bonds and requires roughly 4 × 412 kJ mol$^{-1}$. Options (a) and (b) are duplicated in the source.

**9. Average bond energies for O–O and O=O are 146 and 496 kJ mol$^{-1}$. Find ΔH for H–O–O–H(g) → H–O–H(g) + ½O=O(g).**

(a) −102 kJ
(b) +102 kJ
(c) +350 kJ
(d) +394 kJ

**Answer: (a) −102 kJ** — O–H terms cancel. ΔH = D(O–O) − ½D(O=O) = 146 − 496/2 = 146 − 248 = −102 kJ.

**10. Why does the exothermic reaction C(diamond) → C(graphite), ΔH = −3 kJ mol$^{-1}$, not occur readily?**

(a) the structure of diamond is more stable than that of graphite
(b) diamond has stronger covalent bonds than graphite
(c) the change from diamond to graphite has high activation energy
(d) the density of graphite is less than that of diamond

**Answer: (c) the change from diamond to graphite has high activation energy** — Although the overall change is exothermic, rearranging diamond's strong three-dimensional covalent network requires a very high activation energy.

**Source clarification for MCQ 4:** Yeast fermentation is $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ + energy (ATP). Thus the intended ethanol/ATP option contains a water-for-carbon-dioxide misprint.

## B. Questions for Short Answers

### 5.1. What is the difference between enthalpy and enthalpy change?

**Enthalpy H** is a state property of a system, defined by H = U + PV and often described at this level as its heat content. **Enthalpy change ΔH** is the difference between product and reactant enthalpies for a process: $\Delta H = H_{products} - H_{reactants}$. At constant pressure, ΔH equals the heat absorbed by the system.

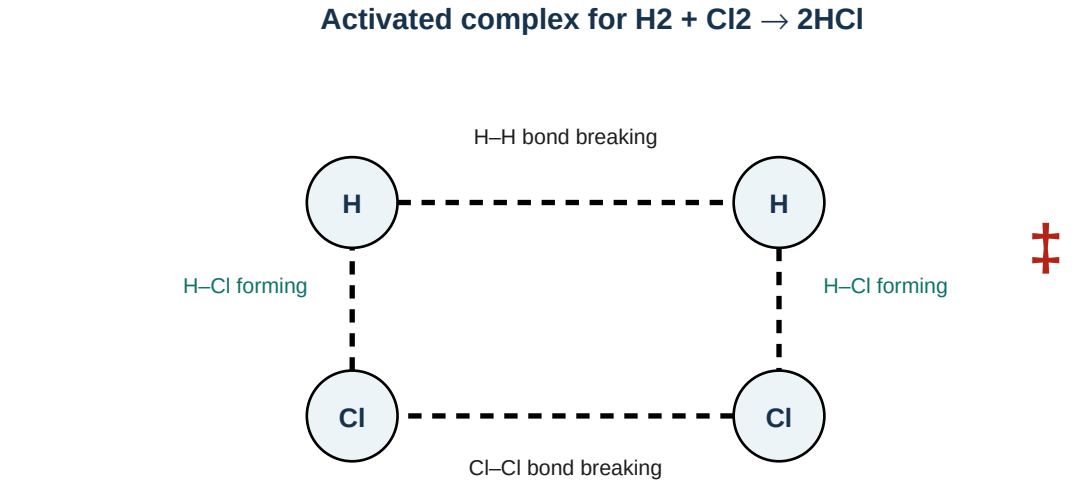
### 5.2. Why is breaking of a bond an endothermic process?

A bonded state is lower in energy because the atoms attract one another. Separating the atoms requires work against this attraction and increases their potential energy. Energy must therefore be absorbed, so bond breaking has a **positive** enthalpy change.

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## 5.3. Depict the transition state for $H_2 + Cl_2 \rightarrow 2HCl$.

**Figure 1 — Original four-centre activated-complex diagram**

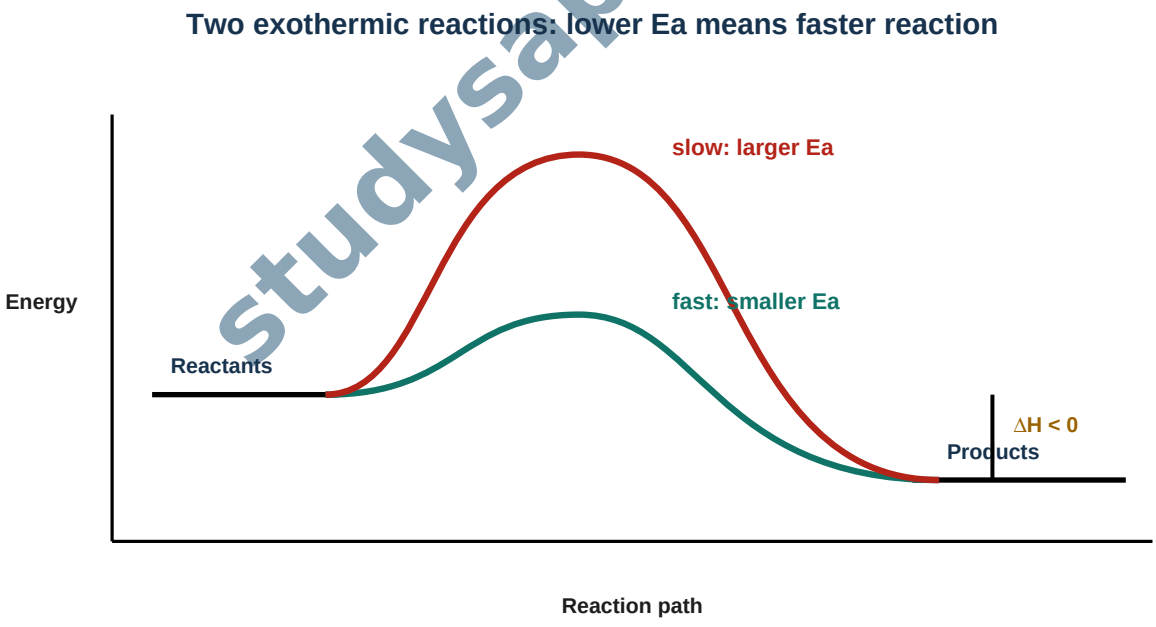



Transition state: highest-energy, short-lived arrangement with partial bonds; it is not an isolable intermediate.

The ‡ symbol marks the transition state. Partial red dashed interactions indicate bonds in the process of breaking or forming; the complex has maximum energy and only an extremely short lifetime.

## 5.4. Draw reaction profiles for two exothermic reactions, one of which moves faster than the other.

**Figure 2 — Original fast/slow exothermic reaction profiles**




Both reactions release the same energy here; activation energy changes rate, not ΔH.

Both products lie below the reactants, so both reactions have $\Delta H < 0$. The faster reaction has the smaller activation-energy hump. A catalyst can produce this effect by providing an alternative lower-barrier pathway.

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### 5.5. What is the role of glycogen in our body?

Glycogen is the short-term storage form of glucose, held mainly in the liver and muscles. Liver glycogen can be broken down to help maintain blood glucose between meals, while muscle glycogen supplies glucose for muscular activity. It is a readily mobilized energy reserve; lipids provide longer-term storage.

## C. Constructed Response Questions

### 5.1. Classify each physical change as exothermic or endothermic.

| Physical change | Classification | Energy reason |
|---|---|---|
| Conversion of hydrated salt into anhydrous salt | Endothermic | Heat is absorbed to remove water of crystallization. |
| Conduction of electricity by metals | Exothermic (when heating is considered) | Electrical energy is partly converted to heat through resistance. |
| Burning of paper | Exothermic | Oxidation releases heat and light. |
| Dissolving ammonium chloride in water | Endothermic | The solution absorbs heat and becomes colder. |
| Vaporizing liquid nitrogen | Endothermic | Liquid nitrogen absorbs latent heat to become gas. |
| Formation of rain from clouds | Exothermic | Water vapour condenses and releases latent heat. |
| Evaporation of dry ice | Endothermic (strictly sublimation) | Solid $CO_2$ absorbs heat to become gas. |
| Dissolving sodium carbonate in water | Exothermic | Hydration and mixing release more energy than is absorbed in separating the solute particles. |

Conduction itself is the transfer of charge; the exothermic classification refers to the accompanying resistive heating. Likewise, dry ice **sublimes** rather than evaporates.

### 5.2. Explain why atmospheric oxygen and nitrogen do not react under normal conditions, but react in lightning to give NO. Why does reaction stop when lightning stops?

$N_2$ contains a very strong N≡N triple bond, and $N_2(g) + O_2(g) \rightarrow 2NO(g)$ is endothermic with a high activation energy. At ordinary temperatures, too few molecules have enough energy for an appreciable rate.

Lightning produces extremely high temperatures and energetic particles, supplying activation energy and enabling NO formation. When lightning ends, the temperature falls and the high-rate formation pathway disappears, so further reaction becomes negligibly slow. Already formed NO need not vanish instantly; it can subsequently react with oxygen to form $NO_2$.

### 5.3. A mixture of natural gas ($CH_4$) and atmospheric oxygen does not react until a burning match is shown, then continues until a reactant is used up. Explain.

At room temperature, $CH_4$ and $O_2$ molecules mostly collide with insufficient energy to overcome the activation barrier, so the mixture is kinetically stable.

The match supplies heat and initiates reactive species, allowing the first combustion events to occur. Combustion is strongly exothermic:

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g) + \text{heat}$$

The released heat raises neighbouring molecules above the activation barrier, so the reaction becomes self-sustaining until fuel or oxygen is exhausted or heat is removed. A fuel–oxygen mixture can therefore be dangerous even before ignition.

## D. Descriptive Questions

### 5.1 Find ΔH for $N_2 + O_2 \rightarrow 2NO$. Bond dissociation energy of $N_2$ = 958.38 kJ $mol^{-1}$, $O_2$ = 498 kJ $mol^{-1}$; bond formation energy of NO = −626 kJ $mol^{-1}$.

**Given:** Break 1 mol N≡N and 1 mol O=O; form 2 mol N=O bonds.

$$\text{Energy absorbed in breaking} = 958.38 + 498 = 1456.38 \text{ kJ mol}^{-1}$$

$$\text{Energy released in forming} = 2 \times (-626) = -1252 \text{ kJ mol}^{-1}$$

$$\Delta H = 1456.38 - 1252$$

**Answer: +204.38 kJ $mol^{-1}$; the reaction is endothermic**

### 5.2. Explain the difference between heat and enthalpy.

| Feature | Heat | Enthalpy |
|---|---|---|
| Meaning | Energy transferred because of a temperature difference | A thermodynamic state property, $H = U + PV$ |
| System property | Heat is energy in transit, not a substance stored in a body | A system has a definite enthalpy for a given state |
| Path or state function | Path-dependent transfer | State function; $\Delta H$ depends on initial and final states |
| Measurement | q, positive when absorbed by the system | $\Delta H = H_{products} - H_{reactants}$ |
| Unit | joule or kilojoule | joule or kilojoule (often quoted per mole of reaction) |

At **constant pressure**, with only expansion work, the heat absorbed by the system equals $\Delta H$. This close relationship is why the terms are sometimes confused, but heat and enthalpy are not conceptually identical.

### 5.3. Explain why formation of a bond is always an exothermic process.

Before bonding, separated atoms have higher potential energy. As attractive interactions draw them together to the equilibrium bond distance, the system reaches a more stable, lower-energy arrangement.

The energy decrease is released to the surroundings, often as heat. Therefore bond formation has a negative enthalpy contribution.

$$H\bullet + \bullet H \rightarrow H\text{–}H \quad \Delta H < 0$$

The reverse process—separating the bonded atoms—requires at least the same energy to be supplied, which is why bond dissociation is endothermic.

### 5.4. Explain the role of lipids in our body.

**Lipids** include fats, oils, waxes and sterols. They are a concentrated long-term energy reserve and are stored in adipose tissue when energy intake exceeds immediate need.

• **Energy:** Stored lipids can be oxidized between meals and during prolonged activity to provide ATP.

• **Insulation and protection:** Adipose tissue reduces heat loss and cushions organs.

• **Cell structure:** Phospholipids and cholesterol are essential components of cell membranes.

• **Regulation and nutrition:** Steroid lipids contribute to hormones, and dietary fat helps absorption of fat-soluble vitamins A, D, E and K.

A balanced amount is essential; excessive stored fat can harm health, while too little dietary fat can interfere with membranes, hormones and vitamin absorption.

### 5.5. Explain activation energy, transition state and aerobic respiration.

**1. Activation energy ($E_a$):** The minimum additional energy reactant particles need to reach the transition state. A larger $E_a$ usually means a slower reaction at a given temperature; a catalyst lowers $E_a$ by offering another pathway.

**2. Transition state:** The highest-energy, very short-lived atomic arrangement along a reaction pathway. Old bonds are partly broken and new bonds partly formed; it cannot normally be isolated.

**3. Aerobic respiration:** The enzyme-controlled oxidation of food in the presence of oxygen to release usable energy. In overall form:

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy (ATP)}$$

The process is exothermic. Cells capture part of the released energy in ATP rather than releasing it all as uncontrolled heat.

## E. Investigative Questions

### 5.1. Why is it essential to cook some food items while others can be eaten without cooking?

Cooking can make food safer and easier to digest. Heat kills many harmful microorganisms, destroys or reduces some natural toxins, denatures proteins, gelatinizes starch and softens fibres. It also develops flavour and can release nutrients from tough structures; meat, dried beans, cassava and many cereals therefore require suitable cooking.

Many ripe fruits, washed salad vegetables and some nuts are naturally soft, digestible and safe to eat raw. Cooking them is unnecessary and can reduce heat-sensitive nutrients such as vitamin C.

The decision therefore depends on **food safety, digestibility, natural toxins, texture and nutrient retention**. Washing and hygienic handling remain important for raw foods.

### 5.2. Why do fireworks look spectacular? What types of chemical compounds undergo reactions during this activity?

Fireworks contain fuel and oxidizing agents that undergo rapid, highly exothermic redox reactions. They release heat, expanding gases, flame, sparks and sound. Carefully engineered delays and compartments control the sequence and pattern.

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| Component / purpose | Examples and effect |
|---|---|
| Oxidizers | Nitrates, chlorates and perchlorates supply oxygen for rapid combustion. |
| Fuels | Charcoal, sulphur, metal powders and other combustibles release energy. |
| Colour producers | Strontium salts: red; sodium salts: yellow; barium compounds: green; copper compounds: blue. |
| Bright flashes and sparks | Magnesium and aluminium burn with intense white light; iron and other particles produce sparks. |
| Black-powder mixture | Commonly $KNO_3$ + charcoal + sulphur; rapid gas production propels and bursts the firework. |

The spectacle therefore combines **activation energy, exothermic oxidation, light emission and gas expansion**. Fireworks are hazardous explosive devices and should only be handled by trained personnel.

## Quick Revision: Energy Sign Checks

**A high activation barrier affects speed; it does not reverse these overall energy signs.**

| Process | Sign of ΔH | Classification | Immediate reason |
|---|---|---|---|
| Break one mole of bonds | + | Endothermic | Energy is required to overcome attraction. |
| Form one mole of bonds | − | Exothermic | A lower-energy bonded state forms. |
| Melt, vaporize or sublime | + | Endothermic | Particle separation increases. |
| Freeze or condense | − | Exothermic | Particles lose energy and attract more closely. |
| Burn a fuel | − | Exothermic | Oxidation releases heat and often light. |
| Neutralize a strong acid with a strong base | − | Exothermic | $H^+ + OH^- \rightarrow H_2O$ releases energy. |

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## Points to Remember

**$\Delta H = H_{products} - H_{reactants}$:** negative for exothermic and positive for endothermic changes.

Bond breaking absorbs energy; bond formation releases energy.

$\Delta H \approx \Sigma D(\text{bonds broken}) - \Sigma D(\text{bonds formed})$, using positive bond-dissociation energies.

**Activation energy** controls rate; ΔH describes overall energy change. A reaction can be exothermic yet extremely slow.

The **transition state** is a maximum-energy partial-bond arrangement, not a stable product or intermediate.

A catalyst increases rate by lowering activation energy without changing ΔH.

Melting, vaporization and sublimation absorb latent heat; freezing and condensation release it.

Aerobic respiration produces $CO_2$, water and ATP; yeast fermentation produces ethanol, $CO_2$ and ATP. Glycogen and lipids are biological energy stores.

### MCQ key at a glance

| Q | Key | Q | Key | Q | Key | Q | Key | Q | Key |
|---|---|---|---|---|---|---|---|---|---|
| 1 | b | 2 | c | 3 | d | 4 | c | 5 | b |
| 6 | a | 7 | d | 8 | d | 9 | a | 10 | c |

### How to attempt this chapter in the paper

• Write the sign convention before calculating: energy in is positive, energy released is negative.
• Count bonds of each type before using bond energies; cancel unchanged bonds only after checking both sides.
• For reaction profiles, label reactants, products, $E_a$, transition state and ΔH; do not confuse barrier height with product energy.
• For physical changes, name latent heat where appropriate and check whether the process is freezing/condensation or melting/vaporization.